

Which Test Answers the Question Asked?

Unit 3 · Topic 3.5 · TODAY'S PROBLEM

A district of 2,400 students had bus ridership $p_0 = 0.40$ before a new route. Before collecting data, the director asks whether the current proportion p has *increased*. An SRS of 180 current students finds 84 riders.

Maya: “Use $H_0 : p = 0.40$ versus $H_a : p > 0.40$ and a one-sample z -test. Check large counts with 72 and 108.”

Theo: “Use $H_a : p \neq 0.40$ because samples can differ either way. Check large counts with the observed 84 and 96.”

Current SRS counts

Group	Bus	Other	Total
Students	84	96	180

FIRST TAKE (5 min, on your sheet)

Does the director want to detect any change, or only more bus riders? Use a phrase from the study.

- DOK 1** (a) Define the population parameter p in context and identify the null value p_0 .
- DOK 2** (b) Compute $\hat{p}$, name the testing procedure, and verify the random, 10 percent, and large-counts conditions using the appropriate values.
- DOK 3** (c) In 3–4 sentences, choose a setup using the director’s word increased. Explain why the count check uses 72 and 108 rather than 84 and 96. State what different question Theo’s alternative would answer.

Optional review: [follow-along](#) (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.

